

Bayesian Techniques in Macroeconomics

Paris-Dauphine & PSL Universities
Master - Quantitative Economics, 2nd year

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CHAPTER 3

Maximum Likelihood Estimation

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HANDOUT OBJECTIVES

- Understanding how to use filters to obtain the sequence of errors;
- Constructing the likelihood function based on the filter's output;
- Optimizing a likelihood function;
- Applying these concepts via Dynare.

Acknowledgement

I thank Ghassane Benmir for providing the Kalman filter codes.

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<https://vermandel.org/btm>.

1. Introduction

There are different way to parametrize a macroeconomic models. These parametrization methods can be listed as follows:

- Calibration;
- Matching Moments;
- Estimation:
 - Maximum Likelihood;
 - Bayesian Estimation;

The calibration strategy usually relies on a [qualitative approach](#). The model that is calibrated is not meant to provide a full description of some economic data, instead, it aims at describing simple economic mechanisms. This is usually addressed through the lens of simple toy models. The problem with the calibration strategy is that it is difficult to choose the right values for the parameters. The difficulty of the calibration strategy can be summarized in a critique by [Pindyck \[2017\]](#) of economic models of climate change as follows: “*Because the modeler has so much freedom in choosing functional forms, parameter values, and other inputs, the model can be used to obtain almost any result one desires (...)*”.

In contrast, the estimation strategy relies on a positive approach: it is aimed at replicating either some specific features of the data (aka '[weak information method](#)'), or all the information offered by the data (aka '[full information method](#)'). This dichotomy is referred to by econometricians as the weak versus full-information methods. Note the first prototype of a DSGE model by [Kydland and Prescott \[1982\]](#) was based on the weak information method in the sense that the model was calibrated to match some moments of the data. Econometricians needed more effort in research to align the model closely with the data. [Taylor and Uhlig \[1990\]](#) and [Fair and Taylor \[1990\]](#) provide some examples of estimated DSGE models using the maximum likelihood method. This handout provides an introduction to the estimation of linear models using full information methods.

2. Filters

Filters are statistical methods which allows to extract relevant information from data. For a macro-econometrician, filtering consists of using the model to extract the sequence of shocks from macroeconomic time series. Note that for a given same sample, there are an infinite number of sequences of shocks and models that can replicate that sample. So the extracted sequence of shocks of one model provide one - among many possible - representation of the data. The key aspect of the estimation is therefore to pick the set of parameters that maximizes the joint density that is the most favored by the data.

2.1 The Kalman filter

The Kalman filter was initially developed for engineering control problems (e.g. Nasa spatial program). In this section, we follow Hamilton's notation and will set up the Kalman filter on general time series model. First, the Kalman filter is linear, so this filter can only be employed for the estimation of linear models. Since most of DSGE models are simulated using a linear approximation, the Kalman filter has become very popular for the estimation of DSGE models.

First, note that when we try to put the model to the data, there are observed and unobserved variables. Think about a RBC model with physical capital, consumption, productivity and investment, an econometrician is able to observe consumption and investment because these are available as macroeconomic time series, while physical capital and productivity are not directly observable.²

The set of unobserved states is given by:

$$x_t = Fx_{t-1} + G\varepsilon_t, \text{ with } \varepsilon_t \sim \mathcal{N}(0, Q) \quad (1)$$

where the covariance matrix of shocks is given by $E(\varepsilon_t, \varepsilon_t') = Q$, $\varepsilon_t \sim N(0, Q)$ is a vector of stochastic disturbances, and F is a full-rank matrix. Note that this transition equation describes how the state vector describes how a set of unobservable state variations evolves over time. Note that the system in [Equation 1](#) is actually the policy rule of our model. Regarding the set of observable variables, the equation reads as:

$$y_t = Hx_t + v_t, \text{ with } v_t \sim \mathcal{N}(0, R) \quad (2)$$

where v_t is a vector of stochastic disturbances interpreted as measurement errors with $v_t \sim N(0, R)$, and H is the matrix that describes how observable variables are related to unobserved ones. This equation will be described in the model as [measurement equations](#).

$$y_t = HFx_{t-1} + HG\varepsilon_t + v_t \quad (3)$$

Note that we don't allow for the stochastic disturbances ε_t and v_t to be correlated (but we could relax this assumption).

Back to the Kalman filter, the latter allows an econometrician to retrieve the values of x_t and v_t that replicate y_t . We assume for now that H , F , G , R and Q are known and time unvarying (without a loss of generality). We provide data Y and matrices H , F and G , and the filter provides the vectors or errors and their (un)conditional moments.

The prediction errors $y_t - E_{t-1}\{y_t\}$ are given by:

$$S_t = y_t - E_{t-1}\{y_t\} = y_t - HE_{t-1}\{x_t\}, \quad (4)$$

²For physical capital, we are not sure exactly at which rate physical capital depreciates. Similarly, the measure of productivity is also subject to endless controversies (think about the Solow's residual).

while the prediction errors variance $\Omega_t = E(S_t S_t')$ is given by:

$$\Omega_t = H E_{t-1} E_{t-1} \{\Sigma_t\} H' + R. \quad (5)$$

The 'best' prediction in the Kalman filter is defined as the optimal mean squared error (MSE) estimator:

$$\min E(y_t - E_{t-1}\{y_t\})(y_t - E_{t-1}\{y_t\})',$$

where K_t represents a closed-form expression of the estimator, it plays a key role in determining how the predictions of the model adjust to surprises (i.e. unexpected shocks ε_t). The formula of K_t reads as follows:

$$K_t = E_{t-1}\{\Sigma_t\} H' \Omega_t^{-1} \quad (6)$$

Next, we can compute the Kalman filter's recursion. We are interested in computing an estimate of x_t given 'surprises' in y_t (surprises are the errors of predictions). Update of the state mean x_t and variance-covariance Σ_t based on the new surprise are given by:

$$x_t = E_{t-1}\{x_t\} + K_t S_t \quad (7)$$

$$\Sigma_t = E_{t-1}\{\Sigma_t\} - K_t H E_{t-1}\{\Sigma_t\} \quad (8)$$

Finally, we can forecast state mean and variance-covariance, based on the current state mean and covariance x_t and Σ_t , as follows:

$$E_t\{x_{t+1}\} = F x_t \quad (9)$$

$$E_t\{\Sigma_{t+1}\} = F \{\Sigma_t\} F' + G Q G' \quad (10)$$

Note that if you need to run the Kalman filter, you need to pre-define H , G , F , R , C , and Q , provide the data y_t , and the filter (based on equations [Equation 4](#)-[Equation 10](#)) provides S_t , Ω_t , x_t , Σ_t , $E_t\{x_{t+1}\}$ and $E_t\{\Sigma_{t+1}\}$. The only last remaining thing that must defined are the initial conditions, $E_0\{x_1\}$ and $E_0\{\Sigma_1\}$, in the recursive equations [7-8](#).

2.2 The Inversion Filter

As an alternative to the Kalman, we can use the inversion filter of [Kollmann \[2013\]](#) that can serve the sequence of errors of any model (either linear or non-linear).

Consider a selection matrix denoted H such that the set of observable is given by $y_t = H x_t$. Note that there as many rows as endogenous variables n_x and as many column as observable variables n_y , this matrix is of all zeros except for row/column observables variables that must include a dummy. Mapping the model y_t to the data Y_t implies that:

$$\begin{aligned} Y_t &= y_t \\ &= H x_t \\ &= H F x_{t-1} + H G \varepsilon_t \end{aligned}$$

We thus solve the equation in ε_t such that the closed-form expression for the contemporaneous realization of shocks:

$$\varepsilon_t = (H G)^{-1} (Y_t - H F x_{t-1}) \quad (11)$$

Note that the error of forecasts, denoted $Y_t - HFx_{t-1}$, is the economic surprise that is rescaled to the model's shock size through the product with the term $(HG)^{-1}$. Therefore using this equation, we can extract the sequence of errors recursively.

To summarize, the inversion filters work by inverting the policy rule to infer the sequence of shock and next compute the current prediction for states variables as follows:

$$\begin{cases} \varepsilon_t = (HG)^{-1} (Y_t - HFx_{t-1}) \\ x_t = Fx_{t-1} + G\varepsilon_t \end{cases},$$

for a sample of size $t = 1 : T$. The usual assumption for initial conditions is to set $x_0 = \bar{x}$ where $\bar{x}$ is the deterministic steady state.

2.3 The particle filter

Particle filters allow the inference of dynamic macroeconomics models, the latter can be non-linear and/or with non-normal errors. We refer to [Fernández-Villaverde and Rubio-Ramírez \[2007\]](#) for a full description of the application to DSGE models.

2.4 A quick discussion

What is the difference between the inversion and the Kalman filter? The Kalman filter is the best linear estimator, so it picks the 'best' sequence of shocks that minimizes the mean squared errors. But if there are as many shocks as observable variables (that is mostly the case for estimated DSGE) then there is only one possible sequence of shock so there is no "best" prediction. However, if there are more shocks than observable variables, then the Kalman is useful to find the 'best' sequence of errors. Note in [Equation 11](#), we can use the inversion filter only if matrix ΛG is square and invertible. Put differently, the inversion filter only works if the number of observable variables is the same as the sequence of errors to be estimated. Given this restriction on the inversion filter, most users still use the Kalman filter to evaluate the sequence of errors of the model. For non-linear models, the particle filter is the best solution to smooth the shocks, however its application is computationally intensive, so the big drawback is the time for the filter to get the sequence of errors.

Now that we are able to get the sequence of errors that replicate some observable variable, the next step is to pick the best set of parameters that replicates the data. The maximum likelihood is the criterion for this.

3. Maximum likelihood

In statistics, maximum likelihood estimation (MLE) is a method of estimating the parameters of a probability distribution by maximizing a likelihood function, so that under the assumed statistical model the observed data is most probable. From a statistical standpoint, a given set of observations are a random sample from an unknown population. The goal of maximum likelihood estimation is to make inferences about the population that is most likely to have generated the sample.

3.1 An illustration with a simple gaussian case

For time series models with gaussian errors and continuous data, let us assume the most simplistic forecasting model of the form:

$$x_t = \varepsilon_t \text{ with } \varepsilon_t \sim \mathcal{N}(\mu, \sigma^2).$$

The forecast error of this model reads as follows:

$$S_t = x_t - E_{t-1}\{x_t\} = \varepsilon_t.$$

Conditional likelihood associated with a realization of x_t can be assessed as the likelihood assigned by its probability density function. Note that if shocks ε_t are gaussian, then the error of predictions are gaussian, $S_t \sim \mathcal{N}(\mu, \sigma^2)$. By assuming normality, we simply assume the shape of our data distribution to conform to the Gaussian bell curve. As we know from statistics, the specific shape and location of our Gaussian distribution come from σ and μ respectively. These two parameters are what define our curve, as we can see when we look at the Probability Density Function (PDF) of the gaussian distribution:³

$$\begin{aligned} P(\varepsilon_t) &= \frac{1}{\sqrt{2\pi V(\varepsilon_t)}} \exp\left(-\frac{(\varepsilon_t - E(\varepsilon_t))^2}{2V(\varepsilon_t)}\right) \\ &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(\varepsilon_t - \mu)^2}{2\sigma^2}\right) \end{aligned}$$

To be technically correct, we can say we are looking for a curve that maximizes the probability of our data given a set of curve parameters. If we want to match a simple gaussian distribution, $\theta = \{\mu, \sigma\}$. We can set up the problem as a conditional probability problem, of which the goal is to maximize the probability of observing our data given θ . Therefore, for a dataset with only one observation (it does not make sense for a continuous probability distribution but it's only for illustrative purpose), we obtain:

$$P(\varepsilon_t|\theta) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(\varepsilon_t - \mu)^2}{2\sigma^2}\right), \text{ with } \theta = \{\mu, \sigma\}. \quad (12)$$

For a dataset of size T , mathematically we can express it as follows:

$$P(x_1, x_2, \dots, x_T|\theta) \quad (13)$$

Because we are dealing with a continuous probability distribution, however, the above notation is technically incorrect, since the probability of observing any set of continuous variables is equal to zero. Conceptually, this makes sense because we can come up with an infinite number of possible variables in the

³Most of this subsection is strongly inspired by <https://towardsdatascience.com/maximum-likelihood-estimation-explained-normal-distribution-6207b322e47f>.

continuous domain, and dividing any given observation by infinity will always lead to a zero probability, regardless of what the observation is.

We need to think in terms of probability density rather than probability. Without going into the technicalities of the difference between the two, we will just state that probability density in the continuous domain is analogous to probability in the discrete domain. Therefore, probability density can be used in this maximization problem. To correct our notation, we will say:

$$\underbrace{P(x_1, x_2, \dots, x_T | \theta)}_{\text{probability}} \rightarrow \underbrace{f(x_1, x_2, \dots, x_n | \theta)}_{\text{density}}$$

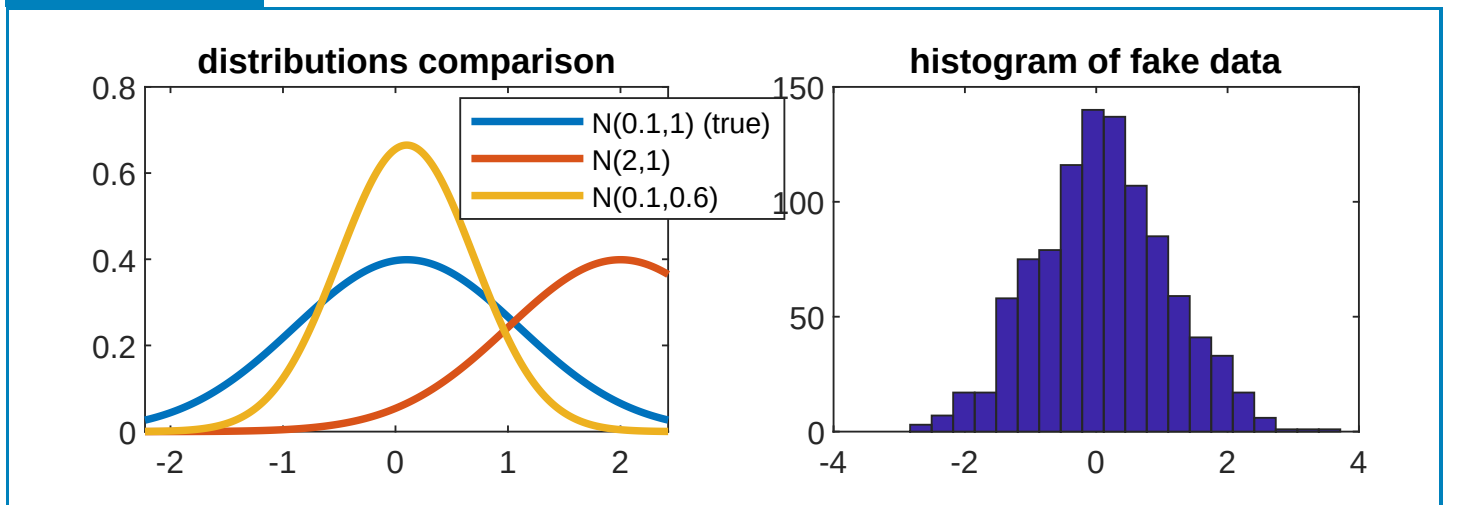
We want to **maximize the probability density of observing our data as a function of θ** . In other words, we want to find μ and σ values such that this probability density term is as high as it can possibly be. We are used to x being the independent variable by convention. But in this case, we are actually treating θ as the independent variable, and we can consider $x_1, x_2, \dots, x_T$ to be a constant, since this is our observed data, which cannot change.

From probability theory, we know that the probability of multiple independent events all happening is termed **joint probability**. We can treat each data point observation as one single event; therefore we can treat the observation of our exact dataset as a series of events, and we can apply joint probability density as follows:

$$\begin{aligned} f(x_1, x_2, \dots, x_T | \theta) &= f(x_1 | \theta) \cdot f(x_2 | \theta) \cdot \dots \cdot f(x_T | \theta) \\ &= \prod_t^T f(x_t | \theta) \end{aligned} \quad (14)$$

To illustrate how the log-likelihood can serve to find the best model, consider the first MATLAB box. We sample some random draws from gaussian distribution $\varepsilon_t \sim N(0.1, 1)$, which is referred to as the true data generating process. Using the sequence of ε_t that we interpret as our sample, we compare alternative distributions drawn respectively from $\theta' = \{2, 1\}$ and $\theta'' = \{0.1, 0.6\}$.

MATLAB box 1 [LK.m](#)



As reported in the left panel, the true gaussian curve in blue (ε_t) is strongly different from the predictions of yellow (θ'') and red (θ') distributions. These theoretical curves can also be compared with the data ε_t in the right panel. Therefore, the goal of the likelihood function as described in [Equation 14](#) is to assess

which of the curve on the left is the closest to data on the right. By calculating the value of the likelihood, we see that $L(Y, \theta) > L(Y, \theta)$ and $L(Y, \theta'')$.

Remember, the goal is to maximize this probability density term by finding the optimal θ . So we are looking for the value of θ that maximizes the probability density, which is also referred to as the maximum likelihood estimator. It can be expressed as follows:

$$\hat{\theta}_{MLE} = \arg \max_{\theta} \Pi_t^T f(x_t | \theta)$$

This problem can be easily solved using optimization routines of MATLAB. However, it's rather typical to apply two tricks to facilitate the manipulation and the resolution of the maximum likelihood problem:

1. Apply logs to linearize (it's easier to code sums than products, it's also easier to calculate gradients on sums than products);
2. Minimize the minus of the likelihood.

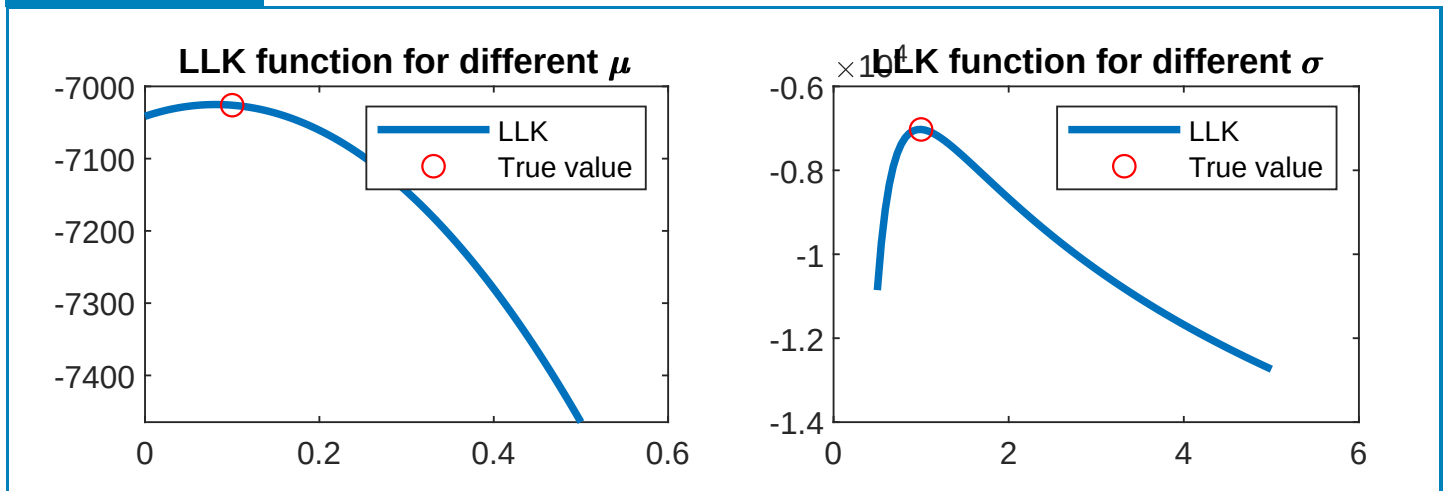
First, we linearize the previous expression:

$$\begin{aligned} \log(\Pi_t^T f(x_t | \theta)) &= \log(f(x_1 | \theta)) + \log(f(x_2 | \theta)) \dots \\ &= \sum_{t=1}^T \log(f(x_t | \theta)) \\ &= \sum_{t=1}^T \left[-\frac{1}{2} \log(2\pi\sigma^2) - \frac{(\varepsilon_t - \mu)^2}{2\sigma^2} \right] \\ &= -\frac{T}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{t=1}^T (\varepsilon_t - \mu)^2 \end{aligned}$$

Finally, in most softwares, it's easier to minimize than maximize, so the maximum likelihood estimator reads as follows:

$$\hat{\theta}_{MLE} = \arg \min_{\{\theta\}} \{-\log(f(x_1, x_2, \dots, x_T | \theta))\} \quad (15)$$

MATLAB box 2 MLE.m



3.2 Maximum likelihood for DSGE models

The previous subsection describes the maximum likelihood estimator with one exogenous variables. However most of macroeconomic models typically includes more than one observable variables. For a n number of observed variables, the errors of predictions of the estimated model follows $S_t \sim \mathcal{N}(\mu_t, \Omega_t)$ with μ_t the mean and Σ_t the covariance of errors of predictions, the general expression of the likelihood function reads as follows:

$$\log L = -\frac{nT}{2} \log(2\pi) - \frac{1}{2} \sum_1^T \log(|\Omega_t|) - \frac{1}{2} \sum_1^T (y_t - E_{t-1}\{y_t\})' \Omega_t^{-1} (y_t - E_{t-1}\{y_t\}). \quad (16)$$

Kalman filter. For an estimation that uses the Kalman filter, the likelihood can be computed by picking the endogenous distribution of errors of predictions in Equation 4 and Equation 5:

$$\log L = -\frac{nT}{2} \log(2\pi) - \frac{1}{2} \sum_1^T \log(|\Omega_t|) - \frac{1}{2} \sum_1^T S_t' \Omega_t^{-1} S_t \quad (17)$$

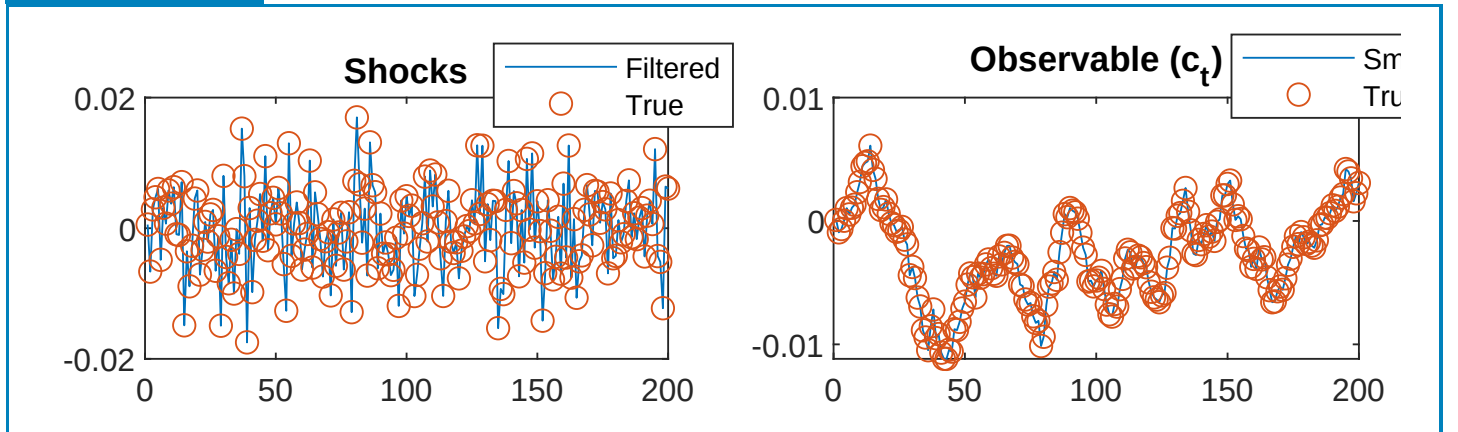
Inversion filter. For an estimation based on the inversion filter to extract the sequence of errors, we have the exact value of the likelihood function (as there is no time-varying uncertainty in the prediction errors stemming from the recursion). The closed-form expression of the errors of predictions are given in Equation 11 with $\varepsilon_t \sim \mathcal{N}(0, Q)$, therefore the likelihood reads as follows:

$$\log L = -\frac{nT}{2} \log(2\pi) - \frac{1}{2} \sum_1^T \log(|Q|) - \frac{1}{2} \sum_1^T \varepsilon_t' Q^{-1} \varepsilon_t. \quad (18)$$

3.3 Illustration with the RBC model

Consider the RBC model with an artificial sample of $T=200$. Recall the we need at least one shock to estimate a model with one observable. Since we only have one shock in the model, our setup only admits one observable variable. This variable will be consumption because it's one of the easy to find macroeconomic time series.

MATLAB box 3 invertRBC.m



Let us first apply the inversion filter. As seen in the MATLAB box, the inversion filter successfully retrieve the sequence of errors that generated the data. The smoothed path of consumption, that we

obtained by feeding the model with the estimated sequences of errors, exactly matches the observed path of consumption.

We can also estimate the model by maximizing the likelihood function in [Equation 18](#) to estimate the variance of errors Q . The estimated variance (given the artificial sample from a random seed) $\hat{Q} = 0.0067$ which is very close to the true value $Q = 0.007$.

Next, we can also apply the Kalman filter to perform the same exercise, we obtain similar results.

4. Maximum Likelihood Estimations with Dynare

4.1 Data transformation

A key problem when performing fit exercises with DSGE models is to map a stationary model with non stationary-data. Macroeconomic time series, like GDP, are better characterized as non-stationary processes that have no tendency to return to a deterministic path as underlined by Nelson and Plosser [1982] and thus requires data transformation to get them stationary.

There are two main components which makes the GDP steadily grow (as underlined by the Solow [1956] model):

1. Technological changes.
2. The population.

First, the first way to remove a non stationary component in the GDP is to divide it by the working population. Indeed, the growth of the population makes the GDP steadily increase at least at the same rate as the population. It is also possible to remove technological changes by rewriting the model in terms of effective labour units as in Solow's model (this step is explained in the following subsection).

Let's first consider the real GDP. You can find for free many macroeconomic time series using DB·NOMICS. It's easy to use (see the DB·NOMICS guide). Once you have downloaded the DB·NOMICS function and have it in your current folder, follow these steps:

1. go to <https://db.nomics.world/>
2. scroll down on the main page, find and click on the OECD database, select the quaterly national account, and find the token (i.e. the key of dbnomics). If you are lazy, the token is:

OECD/QNA/USA.B1_GS1.LNBQRSA.Q

3. Create a Matlab file called "[mydb.m](#)" and use the dbnomics function with the token to import the series into MATLAB.
4. Edit [mydb.m](#) and add the matrix tags for Matlab to read the data. It should look like the following code:

```
1 [my_mat,~,T] = call_dbnomics('OECD/QNA/USA.B1_GS1.LNBQRSA.Q');
```

5. Next, we declare the corresponding vector of quarters, denoted [T](#), converting the dates in matlab format for plotting purposes. Here [gross_y](#) denotes the GDP before transformation (stored in the 2nd column of the data matrix).

```
1 gross_y = my_mat(:,2);
```

6. Recall that in the model, each random variable $\hat{x}_t$ is expressed in terms of detrended percentage deviations from the steady state:

$$\hat{x}_t \simeq \log \left(\frac{X_t}{\bar{X}_0 \gamma^t} \right) \quad (19)$$

where X_t is the non-linear expression of $\hat{x}_t$, $\bar{X}$ is the empirical mean (or the steady state) of X_t and γ^t is the trend component. We transform our sample in the same vein, we first express in terms of log deviations from the trend $\hat{X}_t \simeq \log(X_t/\gamma^t)$, we store the result in a variable called `ln_gap` in Matlab. Then, we remove the steady state, *i.e.* the mean of the sample such that $\hat{x}_t = \hat{x}_t - \mu(\hat{X}_t)$ where $\mu(\cdot)$ is the empirical mean, and add the resulting values to the chart.

To have our in data consistent with the model, (a) we need to apply logs to our data, (b) remove the trend γ^t , (c) remove the mean/steady state:

- a First, we apply the log. Variable `ln_y` is the log of output:

```
1 ln_y = log(gross_y);
```

- b Second we want to remove trend. To do so, we want to know the value of the trend γ that has to be calibrated in the model before the estimation. This parameter is the slope of the trend. Simple linear regression considers only one independent variable using the relation:

$$y_t = \beta_0 + \beta_1 x_t + \epsilon_t$$

where $t = [1, 2, \dots, T]$ is the number of period, y_t is the value of log GDP and ϵ_t is the log deviation from the trend. Here, the trend component is $\beta_1 = \gamma$. We perform the estimation of the trend in Matlab through:

```
1 X = [ones(length(ln_y),1) (1:length(ln_y))'];
2 OLS= X\ln_y;
3 trend_y = X*OLS;
4 gamma = 1+OLS(2);
```

The estimated value of $\gamma = 1.005$, *i.e.* the US economy has experienced a 0.5% quarterly growth rate since 1960. Variable `ln_gap` denotes the detrended output. However, the empirical mean of the output is (still) different from 0.

- c Finally, we remove the empirical mean (which can be interpreted as the steady state of the model):

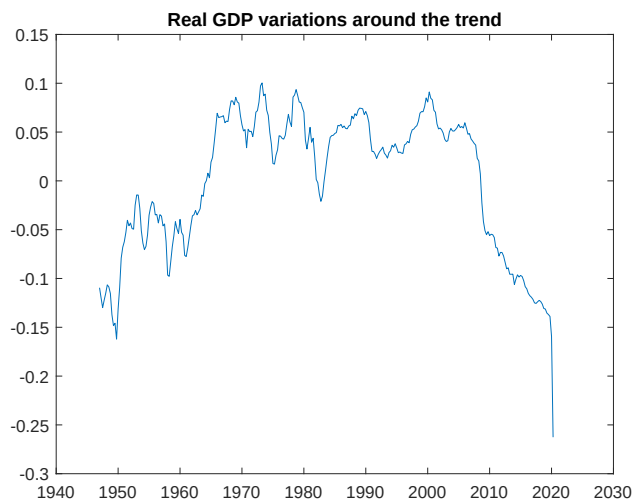


Figure 1: Percentage deviation of the US GDP from its linear trend

```
1 ln_gap = ln_y-trend_y;
2 y = ln_gap-mean(ln_gap);
```

We can plot our result:

```
1 figure;
2 plot(T,ln_y,T,trend_y)
3 title('Real GDP 1940-now')
4 figure;
5 plot(T,y)
6 title('Real GDP variations around the trend')
```

And save the transformed series of output into a matrix file, called `mydata.mat`, which will be read by Dynare right after for the fit exercise.

```
1 save mydata.mat y;
```

Figure 1 plots the transformed GDP, which can be interpreted as a measure of the detrended output gap. It is thus possible to explain the gap between the real GDP and its linear trend through productivity shocks.

Important note: as we are going to call via Dynare the Matlab file "`mydb.m`" we've just created, variable `y` should be declared as an endogenous variable to be estimated in the Dynare file.

4.2 Syntax for log-likelihood estimation

First, let us take back the dynare file of the production economy and include an equation for output.

Second, to estimate using maximum likelihood techniques, first declare the variables of the model which are observed. This is done using the `varobs VAR1 VAR2 VAR3 ;` command, `VARX` must be in the data file a vector of an observed variable, otherwise the estimation won't start. Here, we only have output as observed:

code

```
varobs y;
```

Next, we specify the parameters to be estimated in the `estimated_params;` block. When writing the estimated params section, each parameter to be estimated needs to be given an initial value, and can be given bounds: the syntax to do this is parameter, `initial value [ lower bound, upper bound ]`; where the parts in the square brackets are optional. For the standard errors of the shocks, we use the same syntax, preceding the exogenous variable with `stderr`, like for specifying the shocks. The full estimated params section is:

code

```
estimated_params;  
    stderr    e_a,    0.007,    0,    inf;  
             rho,    0.95,    0,    1;  
end;
```

Finally, we need to call estimation and specify where Dynare can find the observed series. Our observed vector is in the Matlab file `"mydata.mat"`. This is called with:

code

```
estimation(datafile=mydata);
```

4.3 Results from the estimation exercise

Dynare is now solving the optimization problem by finding the value of ρ and the sequence of productivity shocks $\{\eta_t\}_{t=1}^T$ (where T is the size of the sample) which maximize the log-likelihood function. At the beginning of the optimization, the value of the log-likelihood function is `840.5`, while at the end, it's value is `-916.4686337`.

console response

```

Improvement on iteration 10 = 0.000000031
improvement < crit termination
Final value of minus the log posterior (or likelihood):-916.468634
RESULTS FROM MAXIMUM LIKELIHOOD ESTIMATION
parameters
  Estimate s.d. t-stat
rho 0.8362 0.0413 20.2354
standard deviation of shocks
  Estimate s.d. t-stat
e 0.0106 0.0004 24.2567
Total computing time : 0h00m05s

```

According to the estimation output, the optimized parameters are:

$$\rho = 0.8362$$

$$sd(\eta_t) = 0.0106$$

Dynare also gives the standard deviation (the lower is the standard deviation, the better is the identification of the parameter) and the t-stats. Any parameter with the t-stat above 1.96 is statistically significant at the 95% level.

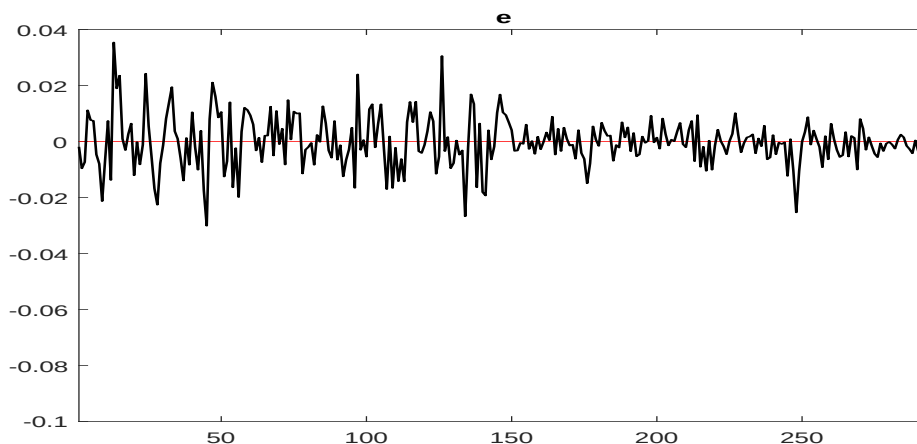


Figure 2: The sequence of estimated smoothed shocks $\{\eta_t\}_{t=1}^T$ over the sample period

Figure 2 plots the estimated sequence of the productivity shock which replicates the best the observed GDP over the sample period.

Finally, we can compare the observed versus the model-generated series in Figure 3. Our model is fairly able to replicate the business cycles of the US economy over the sample period.

The estimation results are saved in the `oo_` structure in the workspace. Some details are given in Table 1.



Figure 3: Observed series (black) and model-generated series (red) which use the estimated sequence of the productivity shock drawn in [Figure 2](#)

Stored in	Description
oo_.SmoothedVariables	Estimated values of each endogenous variable for all time periods.
oo_.SmoothedShocks	Estimated values of each shock for all time periods.

Table 1: Matlab's workspace after a likelihood estimation

It is possible to reverse the detrending method in order to get estimated data back to their raw values. This operation may look useless at first sight, but it allows us to plot and analyse the unobserved path of capital and productivity over the business cycles. First, we can plot detrended series:

code

```
load mydata;
figure
subplot(3,1,1);
plot(T,oo_.SmoothedVariables.y)
title('Output')
subplot(3,1,2);
plot(T,oo_.SmoothedVariables.k)
title('Capital')
subplot(3,1,3);
plot(T,oo_.SmoothedVariables.a)
title('Productivity')
```

The result from the previous code is drawn in [Figure 4](#). From this figure, it is possible to explain the persistence of negative output gaps since the financial crisis by an important decline in physical capital stock. Another important feature of the business cycles is the variance of productivity shocks which dropped strongly after 1980 (by the way do you know what have made the business cycles different after 1980?)

To reverse the detrending method, we just need to add up the trend component and apply the expo-

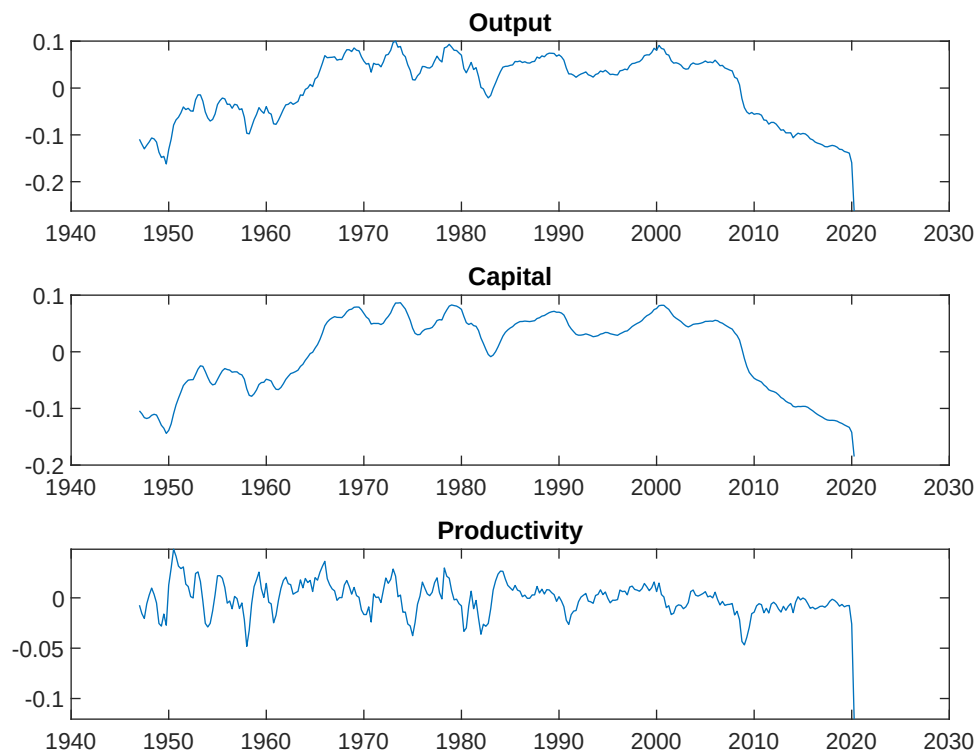


Figure 4: Model-generated output, capital and productivity (using the estimated model and sequence of shocks).

nential function `exp(·)` to reverse the log. We don't need to plot the productivity shock as it is assumed to be detrended and stationary.

code

```
% REVERSING
figure;
subplot(2,1,1);
plot(T,exp(trend_y+oo_.SmoothedVariables.y))
title('Output')
subplot(2,1,2);
plot(T,exp(trend_y+oo_.SmoothedVariables.k))
title('Capital')
```

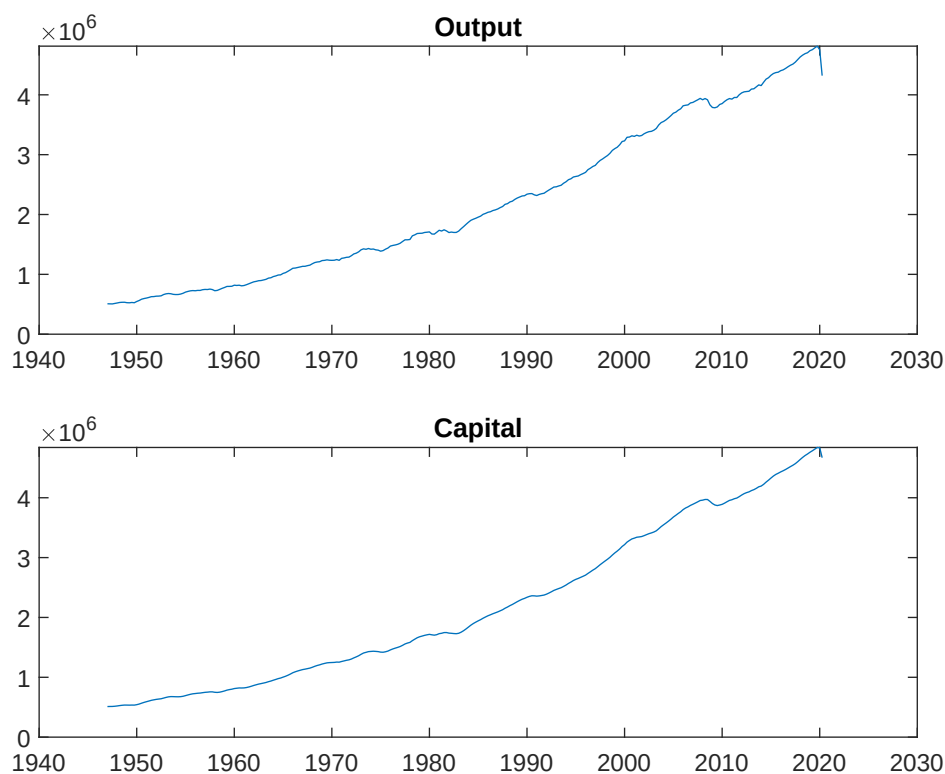


Figure 5: GDP and physical capital generated by the model after reversing the detrending method

Exercise 1: Second moment statistics and IRFs of the estimated model

1. Compute using Matlab's function `std(.)` the standard deviation of the estimated sequence of the productivity shock. Do you find the same result as Dynare?
2. Compare the standard deviation of the model with the one observed in the data. Is your model empirically relevant?
3. In your opinion, why is it important to have second order statistics relevant with the data?
4. Plot the IRFs of the estimated model.

Exercise 2: Estimating the model with output and consumption

Consider now adding another time serie in the model: consumption. It is thus possible to get directly from dbnomics.

To avoid stochastic singularity issue, you must have the same number of shocks (or more) than observables. Estimation is ok as long as estimated parameters (including shock variances) are identified (i.e. different from 0 and not estimated on their lower/upper bound). To estimate consumption, we need to add one more shock to the model. To increase demand shock effects, we also include investment I_t in the GDP.

Let's consider a public spending shock in the GDP:

$$Y_t = C_t + I_t + G_t$$

Public spending G_t is financed by a tax T_{jt} charged to all households populating our economy, assuming a balanced government budget over the cycles:

$$G_t = \int_0^1 T_{jt} dj$$

This tax affects the budget constraint of the household:

$$C_{jt} + K_{jt} + T_{jt} = A_t K_{t-1}^\alpha + (1 - \delta) K_{t-1}$$

Spending are exogenously determined as a constant fraction of the GDP ($\bar{G} = g^y \bar{Y}$):

$$G_t = \bar{G} \varepsilon_t^G \text{ with } \log(\varepsilon_t^G) = \rho_G \log(\varepsilon_{t-1}^G) + \eta_t^G$$

our model is now summarized by:

$$\hat{c}_t = -\beta \alpha \bar{k}^{\alpha-1} (\mathbb{E}_t \hat{a}_{t+1} - (1 - \alpha) \hat{k}_t) + \beta [\alpha \bar{k}^{\alpha-1} + (1 - \delta)] \mathbb{E}_t \hat{c}_{t+1} \quad (20)$$

$$\hat{y}_t = \hat{a}_t + \alpha \hat{k}_{t-1} \quad (21)$$

$$\bar{y} \hat{y}_t = \bar{c} \hat{c}_t + \bar{i} \hat{i}_t + g^y \bar{y} \hat{g}_t \quad (22)$$

$$\delta \hat{i}_t = \hat{k}_t - (1 - \delta) \hat{k}_{t-1} \quad (23)$$

$$\hat{a}_t = \rho_A \hat{a}_{t-1} + \eta_t^A \quad (24)$$

$$\hat{g}_t = \rho_G \hat{g}_{t-1} + \eta_t^G \quad (25)$$

with $g^y = 0.2$, *i.e.* spending shocks accounts for 20% of output variations. The steady state thus reads as follows:

$$\begin{aligned} \bar{k} &= \left(\left[\frac{1}{\beta} - (1 - \delta) \right] \frac{1}{\alpha} \right)^{1/(\alpha-1)} \\ \bar{i} &= \delta \bar{k} \\ \bar{y} &= \bar{k}^\alpha \\ \bar{c} &= (1 - g^y) \bar{y} - \bar{i} \end{aligned}$$

Our economy is characterized by a set of six variables $\{\hat{c}_t, \hat{k}_t, \hat{y}_t, \hat{i}_t, \hat{a}_t, \hat{g}_t\}$.

1. Code the new model.

2. Estimate the new model with consumption and output, estimate the variance of the spending shock as well as its AR(1) term.
3. Analyse the variance decomposition. Which shock is driving/explaining the business cycles for the US economy?
4. Is the slope of the trend different for consumption and the GDP? (In the next exercise, we will develop a method to estimate one single trend for both consumption and GDP).
5. Let's try to develop and compare an alternative shock which could explain the US business cycles. Include the following preference shock to the model:

$$\max_{C_{jt}, K_{jt}} \mathbb{E}_t \sum_{\tau=0}^{+\infty} \beta^\tau \varepsilon_{t+\tau}^C \log(C_{jt+\tau})$$

- (a) How the Euler equation is affected by this shock?
- (b) Estimate this new model with its AR(1) terms defined as:

$$\log(\varepsilon_t^C) = \rho_C \log(\varepsilon_{t-1}^C) + \eta_t^C$$

- (c) Which model is better given the data? (compare the model with government spending shock, preference shock and both shocks).

Exercise 3: Estimating deep parameters

Consider in this exercise the simple model without consumption/investment. It is possible to estimate parameters such as β , α and δ . However, these parameters affect the steady state of the model. Indeed, recall that:

$$\bar{k} = \left(\left[\frac{1}{\beta} - (1 - \delta) \right] \frac{1}{\alpha} \right)^{1/(\alpha-1)}$$

$$\bar{c} = \bar{k}^\alpha + \delta \bar{k}$$

$$\bar{y} = \bar{k}^\alpha$$

Each time β , α and δ are changed to maximize the log-likelihood function, the steady state should be updated accordingly. It is thus easy in Dynare to update the steady of the model. In `model(linear);`'s block, add:

code

```
#Kss = ((1/beta-(1-delta))/alpha)^(1/(alpha-1));
#Css = Kss^alpha - delta*Kss;
```

At the beginning of your Dynare file, you shall remove Kss and Css from the parameter's block `parameters ...;` to make dynare update these parameters each iteration.

Any parameter in model's block preceded by the sign "#" is updated by Dynare each iteration of the optimization algorithm employed in the fit exercise.

1. Estimate the value δ and look at the t-stat. Is this parameter statistically significant? From this result, try to explain why δ is not well identified.
2. Do you find a similar result for β , α ? Given the data, is your fit better with or without these parameters?
3. [Smets and Wouters \[2007\]](#) obtain a well identified estimation of α and β for the US economy. Do you find such a result with that simple RBC model?

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